

Financial Econometrics

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Dynamic Factor Models

- 1 Introduction
- 2 Dynamic Conditional Beta model
- 3 Cholesky decomposition

The rejection of the CAPM, and its multi-factors extensions (APT), in empirical studies can be due to the fact that the regression parameters are **changing over time**.

It is therefore important to develop simple statistical approaches capturing all the **dynamic aspects** of the financial series

Dynamic factor model

A **dynamic** factor model writes

$$\mathbf{y}_t = \mathbf{c}_t + \mathbf{B}_t \mathbf{f}_t + \mathbf{u}_t$$

where

- $\mathbf{f}_t \in \mathbb{R}^K$ is the vector of factors
- $\mathbf{c}_t$ and $\mathbf{B}_t$ (loading matrix) are matrices of sizes $N \times 1$ and $N \times K$,
- $\mathbf{u}_t \in \mathbb{R}^N$ is the vector of idiosyncratic components, which is uncorrelated with the factors :

$$E(\mathbf{u}_t | \mathbf{f}_s) = \mathbf{0}, \quad E(\mathbf{u}_t \mathbf{u}_s' | \mathbf{f}_s) = \mathbf{\Sigma}, \quad t, s \in \mathbb{Z}.$$

- $\mathbf{\Sigma}$ is a semi-positive definite.

Drawbacks of some methods used for handling models with time-varying parameters

- **Deterministic modeling of the parameters**

Such models use splines , trends..

The implications in terms of statistical inference are unclear, and extrapolation may lead to inconsistent results.

Drawbacks of some methods used for handling models with time-varying parameters

- **State space models**

Parameters are treated as space variables. For instance, for a single-factor model (e.g. the CAPM)

$$\begin{aligned}\mathbf{y}_t &= \boldsymbol{\beta}_t f_t + \mathbf{u}_t, \\ \boldsymbol{\beta}_t &= \rho \boldsymbol{\beta}_{t-1} + \mathbf{v}_t, \quad \boldsymbol{\beta}_t \in \mathbb{R}^N\end{aligned}$$

This approach can be efficient and statistically founded.

But requiring a specific stochastic process for the coefficient $\boldsymbol{\beta}$ may entail considerable numerical difficulties.

Moreover, interpretations are difficult in this framework.

Drawbacks of some methods used for handling models with time-varying parameters

- **Rolling window regression (a)**

Consists in performing a rolling regression (i.e. regressions with a rolling time window), with subsamples of the original full sample.

- For example, regressions using windows with a size of 50 each, i.e. from 1 :50, then from 51 :100 etc.
- Another approach would be to apply overlapping windows with a size of 50 each. So for example using 1 :50, then 41 :90 etc.

As a result one gets a time series of regression coefficients, which can be analyzed. This approach is very popular among practitioners (and also in the economics and finance literature).

Drawbacks of some methods used for handling models with time-varying parameters

- **Rolling window regression (b)**

However, this approach generally employs an **inconsistent set of assumptions**.

- If the model is **stationary**, using incomplete data for estimating constant 'betas' entails **efficiency loss**.
- If the model is **non stationary**, some of the estimators may be **inconsistent**.
- This approach is unable to capture non-constant betas in a stationary framework (see later)

References I

- [1] Engle R.F. (2016)
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- [2] Darolles S., Francq C. and S. Laurent (2018)
Asymptotics of Cholesky GARCH models and time-varying conditional betas. *Journal of Econometrics* 204, 223–247.

- 1 Introduction
- 2 **Dynamic Conditional Beta model**
 - Model
 - Estimation
 - Illustration
- 3 Cholesky decomposition

Dynamic Conditional Beta model

Recently, Engle (2016) proposed a model called **Dynamic Conditional Beta (DCB)** to obtain time-varying slope coefficients.

In this model the factors are **observable**.

Assuming **joint conditional normality** of the variables appearing in the model (factors and dependent variables), the **time-varying slope coefficients (betas)** of the factors can be recovered using an estimate of the **full conditional covariance matrix**.

In applications, one can rely on a **Dynamic Conditional Correlation (DCC)** model to obtain the conditional betas (e.g. for the dynamic Fama-French three factor model).

Model

Assume that the joint **conditional distribution** of the **returns and factors** is Gaussian :

$$\mathcal{L}\left(\begin{pmatrix} \mathbf{y}_t \\ \mathbf{f}_t \end{pmatrix} \middle| I_{t-1}\right) = \mathcal{N}\left(\begin{pmatrix} \mathbf{c}_t \\ \mathbf{0}_K \end{pmatrix}, \quad \mathbf{Q}_t := \begin{pmatrix} \boldsymbol{\Omega}_t & \mathbf{H}_t \\ \mathbf{H}_t' & \boldsymbol{\Lambda}_t \end{pmatrix}\right)$$

where I_{t-1} is the information on all past variables, $\mathbf{c}_t \in \mathbb{R}^N$ and $\mathbf{Q}_t$ is a positive definite matrix.

► Property of Gaussian vectors

The distribution of $\mathbf{y}_t$ conditional on I_{t-1} and $\mathbf{f}_t$ is

$$\mathcal{N}(\mathbf{c}_t + \mathbf{H}_t \boldsymbol{\Lambda}_t^{-1} \mathbf{f}_t, \quad \boldsymbol{\Omega}_t - \mathbf{H}_t \boldsymbol{\Lambda}_t^{-1} \mathbf{H}_t')$$

The model entails the following conditional moments restrictions :

$$\begin{aligned} E(\mathbf{y}_t | \mathbf{f}_t, I_{t-1}) &= \mathbf{c}_t + \mathbf{H}_t \boldsymbol{\Lambda}_t^{-1} \mathbf{f}_t \\ V(\mathbf{y}_t | \mathbf{f}_t, I_{t-1}) &= \boldsymbol{\Omega}_t - \mathbf{H}_t \boldsymbol{\Lambda}_t^{-1} \mathbf{H}_t' := \boldsymbol{\Sigma}_t \end{aligned}$$

Factor representation

Thus

$$\mathbf{y}_t = \mathbf{c}_t + \underbrace{\mathbf{H}_t \boldsymbol{\Lambda}_t^{-1}}_{\mathbf{B}_t} \mathbf{f}_t + \mathbf{u}_t$$

where the errors are conditionally Gaussian with

$$E(\mathbf{u}_t | \mathbf{f}_t, I_{t-1}) = \mathbf{0}, \quad V(\mathbf{u}_t | \mathbf{f}_t, I_{t-1}) = \boldsymbol{\Sigma}_t.$$

The betas from this expression are called **Dynamic Conditional Betas** :

$$y_{it} = c_{it} + \sum_{j=1}^K \beta_{ij,t} f_{jt} + u_{it}$$

where $\mathbf{B}_t = (\beta_{ij,t})_{i,j}$.

By construction : $\mathbf{B}_t$ belongs to the past of $\mathbf{y}_t$ and $\mathbf{f}_t$.

Case of a single asset

When $N = 1$,

$$y_t = c_t + \underbrace{\boldsymbol{\beta}'_t}_{1 \times K} \underbrace{\mathbf{f}_t}_{K \times 1} + u_t$$

and

$$\boldsymbol{\beta}'_t = \text{Cov}(y_t, \mathbf{f}_t | I_{t-1}) \{V(\mathbf{f}_t | I_{t-1})\}^{-1}$$

Remarks on the assumptions

The **normality assumption** is clearly restrictive but can be generalized. For example, if the joint conditional distribution is in the **elliptical family** of distributions, then the conditional distribution of $\mathbf{y}_t$ given $\mathbf{f}_t$ is elliptical.

Without any assumption on the conditional distribution, the factor model can be interpreted as the **conditional linear projection**. In this case the error term has a non-normal distribution.

ML estimation

Observations : $\mathbf{y}_1, \dots, \mathbf{y}_n$ and $\mathbf{f}_1, \dots, \mathbf{f}_n$. Let $\mathbf{x}_t = \begin{pmatrix} \mathbf{y}_t \\ \mathbf{f}_t \end{pmatrix}$.

Suppose that the different matrices are parameterized by a parameter $\boldsymbol{\theta}$:

$$\mathbf{c}_t = \mathbf{c}_t(\boldsymbol{\theta}), \quad \mathbf{H}_t = \mathbf{H}_t(\boldsymbol{\theta}), \quad \boldsymbol{\Lambda}_t = \boldsymbol{\Lambda}_t(\boldsymbol{\theta}), \quad \boldsymbol{\Omega}_t = \boldsymbol{\Omega}_t(\boldsymbol{\theta}).$$

Then, ignoring irrelevant constants, the **log-likelihood** has the form

$$L_n(\boldsymbol{\theta}) = -\frac{1}{2} \sum_{t=1}^n \log |\mathbf{Q}_t(\boldsymbol{\theta})| + \{\mathbf{x}_t - \mathbf{m}_t(\boldsymbol{\theta})\}' \mathbf{Q}_t(\boldsymbol{\theta})^{-1} \{\mathbf{x}_t - \mathbf{m}_t(\boldsymbol{\theta})\}$$

If $\hat{\boldsymbol{\theta}}$ denotes the MLE of $\boldsymbol{\theta}_0$ (the true parameter value), an **estimator of the dynamic beta** coefficients is then

$$\hat{\mathbf{B}}_t = \mathbf{H}_t(\hat{\boldsymbol{\theta}}) \boldsymbol{\Lambda}_t^{-1}(\hat{\boldsymbol{\theta}}).$$

Parametrization of the dynamics

For a **single time series** of returns y_t ($N=1$), a standard modeling is to specify the dynamics of the **volatility** via a GARCH (Generalized Autoregressive Conditionally Heteroskedastic) model of the form

$$y_t = \sigma_t(\boldsymbol{\theta}_0)\eta_t$$

where the volatility σ_t is a parametric function of the past returns, $\boldsymbol{\theta}_0 \in \mathbb{R}^d$ and (η_t) is an i.i.d. sequence of variables, with $E\eta_t = 0$ and $V(\eta_t) = 1$:

$$\sigma_t(\boldsymbol{\theta}_0) = \sigma(y_{t-1}, y_{t-2}, \dots; \boldsymbol{\theta}_0) > 0.$$

Parametrization of the dynamics

The **standard model** is the GARCH(1,1),

$$\begin{cases} y_t &= \sigma_t(\boldsymbol{\theta}_0)\eta_t, \\ \sigma_t^2(\boldsymbol{\theta}_0) &= \omega_0 + \alpha_0 y_{t-1}^2 + \beta_0 \sigma_{t-1}^2(\boldsymbol{\theta}_0), \end{cases}$$

$\boldsymbol{\theta}_0 = (\omega_0, \alpha_0, \beta_0)'$, with $\omega_0 > 0$, and $\alpha_0, \beta_0 \geq 0$.

The strict stationarity condition is $E \log(\alpha_0 \eta_t^2 + \beta_0) < 0$ (assuming the expectation exists).

Specification of the conditional variance

A natural generalization is to consider the model (with $\mathbf{c}_t = \mathbf{0}$)

$$\begin{pmatrix} \mathbf{y}_t \\ \mathbf{f}_t \end{pmatrix} = \mathbf{Q}_t^{1/2}(\boldsymbol{\theta}_0) \boldsymbol{\eta}_t$$

where $\mathbf{Q}_t$ is a "volatility matrix" and $\boldsymbol{\eta}_t \stackrel{i.i.d.}{\sim} (\mathbf{0}, \mathbf{I})$.

A multivariate GARCH model can be used : Engle (2016) uses a **DCC-GARCH** model.

► Examples of MGARCH

The DCC approach relies on the following decomposition of $\mathbf{Q}_t$:

$$\mathbf{Q}_t(\boldsymbol{\theta}_0) = \mathbf{D}_t(\boldsymbol{\theta}_0) \mathbf{R}_t(\boldsymbol{\theta}_0) \mathbf{D}_t(\boldsymbol{\theta}_0)$$

where $\mathbf{D}_t(\boldsymbol{\theta}_0)$ is a diagonal matrix with the **conditional volatilities** of $\mathbf{x}_t$, $\mathbf{R}_t(\boldsymbol{\theta}_0)$ is its **conditional correlation matrix**.

Note that, even with a **CCC-GARCH** model ($\mathbf{R}_t = \mathbf{R}$), the **betas are not constant**.

Dynamic versus constant betas ?

The model

$$\mathbf{y}_t = \mathbf{c}_t + \mathbf{B}_t \mathbf{f}_t + \mathbf{u}_t, \quad \text{with } \mathbf{B}_t \in I_{t-1}, \quad E(\mathbf{u}_t | \mathbf{f}_t, I_{t-1}) = \mathbf{0}$$

is **not incompatible** with a static model (with **constant beta**) of the form

$$\mathbf{y}_t = \mathbf{c} + \mathbf{B} \mathbf{f}_t + \mathbf{v}_t, \quad \text{with } E(\mathbf{v}_t | \mathbf{f}_t) = \mathbf{0}.$$

The latter regression exists whenever $(\mathbf{y}_t, \mathbf{f}_t)$ is stationary with finite variance.

Therefore, the procedure consisting in computing rolling window regressions to see if the estimated beta vary is inconsistent.

Testing for constant betas

Having estimated the dynamic model (1st step), it is suggested in Engle (2016) to consider (2nd step) the artificial regression (written for $K = 1$ factor and $N = 1$) :

$$y_t = \mu_t + \beta_t(\hat{\theta})\gamma f_t + \beta f_t + \epsilon_t$$

and consider Wald test of the null assumption of **constant beta** :

$$H_0 : \gamma = 0$$

or of the null assumption of **dynamic beta** :

$$H_0^* : \beta = 0.$$

Application on an industry portfolio

12 industry portfolios (Equipment, Finance, Energy, Telecoms, ...)

Data cover the period 1963-2011 ($n = 12,000$ daily observations)

For each industry, the **benchmark Fama-French** model is given by :

$$z_{it} = c_i + B_{i1}z_t^{(m)} + B_{i2}\text{SMB}_t + B_{i3}\text{HML}_t + \sqrt{h_{it}}u_{it}, \quad t = 1, \dots, n$$

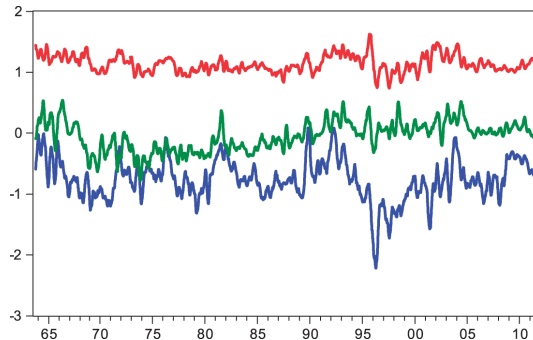
The DCC model is used to estimate the **conditional covariance matrix** of the 3 factors and the portfolio.

Conditional betas for Business Equipment

Computers, softwares, electronic equipment

The **Market** beta is red, the **HML** beta is blue and the **SMB** beta is green. Buseq has a negative beta on HML in the 90s. This sector trades like a growth stock with time varying intensity.

Figure 1. Smoothed DCBs are computed for Business Equipment from the nested DCB. The Market beta is red, the HML beta ...

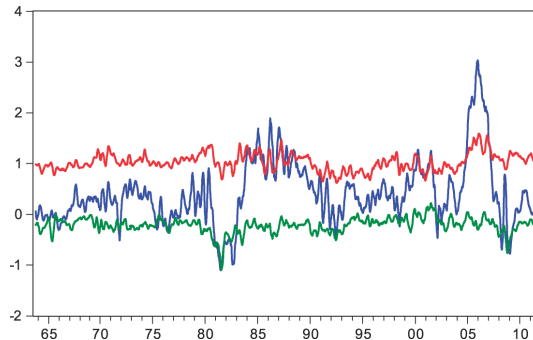


Conditional betas for Energy

Oil, Gas, and Coal Extraction and Products

The **Market** beta is red, the **HML** beta is blue and the **SMB** beta is green. This sector trades like a value stock in 1982-90 and 2003-07, and like a growth stock in 1980-82.

Figure 2. Smoothed DCBs for Energy are computed from the nested DCB. The market beta is red, the HML beta is blue and ...

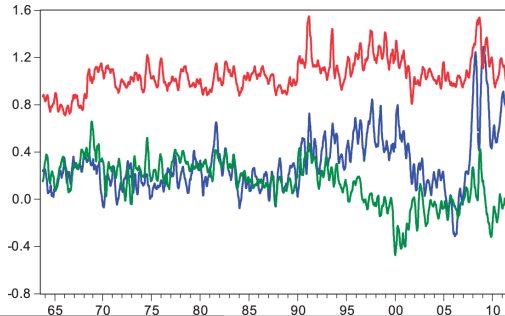


Conditional betas for Money

Finance

HML beta : this sector trades like a growth stock in 2004-07 with a sudden change during the financial crisis. **Market** beta : increases during the crisis. **SMB** beta : becomes negative after 1990 (the sector trades like a large cap stock).

Figure 3. Smoothed DCBs for Money are computed from the nested DCB. The market beta is red, the HML beta is blue and ...



Merits and drawbacks

The specificity of this approach is to **deduce the conditional betas** from an estimator of the matrix $\mathbf{Q}_t(\boldsymbol{\theta}_0)$.

This approach is relatively easy to implement, but the asymptotic properties of the ML estimation of the DCC are unknown.

Moreover, testing and imposing the constancy of the conditional betas can be complicated.

Furthermore, identifying the **variables determining the evolution of the betas** is impossible : conditional betas are **not modeled directly** but recovered afterwards by a non-linear transformation of the elements of the estimated conditional covariance matrix.

- 1 Introduction
- 2 Dynamic Conditional Beta model
- 3 Cholesky decomposition**
 - Model
 - Parametrization and estimation
 - Illustration

Cholesky decomposition of the conditional variance matrix

Introduced by Pourahmadi (1999) for longitudinal data. Extended to time series by Darolles, Francq and Laurent (2018).

Consider a **single time-series** y_t and a vector of K factors $\mathbf{f}_t$. Let

$$\mathbf{z}_t := \begin{pmatrix} \mathbf{f}_t \\ y_t \end{pmatrix} \in \mathbb{R}^{K+1}.$$

Suppose that

$$\mathbf{z}_t = \mathbf{Q}_t^{1/2} \boldsymbol{\eta}_t$$

where $(\boldsymbol{\eta}_t)$ is a sequence of iid random vectors with zero mean and identity covariance matrix, and $\mathbf{Q}_t$ is a positive-definite covariance matrix, function of the past values of $\mathbf{z}_t$.

Cholesky decomposition of the conditional variance matrix

This approach aims at decomposing the conditional covariance matrix of $\mathbf{z}_t$ as

$$V(\mathbf{z}_t | I_{t-1}) = \mathbf{Q}_t = \begin{pmatrix} \mathbf{\Lambda}_t & \mathbf{H}_t' \\ \mathbf{H}_t & \mathbf{\Omega}_t \end{pmatrix} = \mathbf{L}_t \mathbf{G}_t \mathbf{L}_t',$$

where $\mathbf{L}_t, \mathbf{G}_t \in I_{t-1}$ and

- $\mathbf{L}_t$ is a lower-triangular matrix with 1 on the diagonal
- $\mathbf{G}_t$ is a diagonal matrix of variance of **orthogonal** factors :

$$\mathbf{G}_t = \text{diag}(g_{1,t}, \dots, g_{K+1,t}), \quad g_{k,t} = V_{t-1}(v_{k,t}), \quad \text{Cov}_{t-1}(v_{k,t}, v_{j,t}) = 0.$$

Consequence of the Cholesky decomposition of the conditional variance matrix

$$V(\mathbf{z}_t|I_{t-1}) = \mathbf{L}_t \mathbf{G}_t \mathbf{L}_t' \quad \Rightarrow \quad V(\mathbf{B}_t \mathbf{z}_t | I_{t-1}) = \mathbf{G}_t,$$

where

$$\mathbf{B}_t = \mathbf{L}_t^{-1} =: (-\beta_{i,j,t})$$

is a lower-triangular matrix with 1 on the diagonal.

In particular, for the last row of $\mathbf{B}_t \mathbf{z}_t$ we have,

$$V_{t-1} \left(- \sum_{j=1}^K \beta_{K+1,j,t} f_{jt} + y_t \right) = V_{t-1}(v_{K+1,t}).$$

We will construct a sequence of orthogonal factors such that :

$$y_t = \sum_{j=1}^K \beta_{K+1,j,t} f_{jt} + v_{K+1,t}$$

Cholesky decomposition of the conditional variance matrix

Let $v_{1t} = f_{1t}$ and consider the **theoretical regression** of f_{2t} on v_{1t} conditional on I_{t-1} : one can write

$$f_{2t} = \ell_{21,t} v_{1t} + v_{2t} = \beta_{21,t} f_{1t} + v_{2t}, \quad (\beta_{21,t} = \ell_{21,t})$$

where $\ell_{21,t} \in I_{t-1}$ and v_{2t} is orthogonal to v_{1t} conditional on I_{t-1}

$$\Rightarrow \quad \ell_{21,t} = \frac{\text{Cov}_{t-1}(f_{1t}, f_{2t})}{V_{t-1}(f_{1t})}$$

Cholesky decomposition of the conditional variance matrix

More generally, for $i = 2, \dots, K$,

$$f_{it} = \sum_{j=1}^{i-1} \ell_{ij,t} v_{jt} + v_{it} = \sum_{j=1}^{i-1} \beta_{ij,t} f_{jt} + v_{it}$$

where v_{it} is conditionally orthogonal to v_{jt} ($1 \leq j \leq i-1$), and thus to f_{jt} ($1 \leq j \leq i-1$).

Finally, consider the regression of y_t on the K factors

$$y_t = \sum_{j=1}^K \ell_{K+1,j,t} v_{jt} + v_{K+1,t} = \sum_{j=1}^K \beta_{K+1,j,t} f_{jt} + v_{K+1,t}$$

where $v_{K+1,t}$ is conditionally orthogonal to the factors and $\ell_{ij,t} \in I_{t-1}$.

Cholesky decomposition of the conditional variance matrix

In matrix form

$$\mathbf{z}_t = \begin{pmatrix} \mathbf{f}_t \\ y_t \end{pmatrix} = \mathbf{L}_t \mathbf{v}_t \quad \Leftrightarrow \quad \mathbf{v}_t = \mathbf{B}_t \mathbf{z}_t$$

where $\mathbf{v}_t = (v_{1t}, \dots, v_{K+1,t})'$ and $\mathbf{L}_t$ and $\mathbf{B}_t$ are a lower-triangular matrices with 1 on the diagonal.

Note that

$$\mathbf{z}_t = \mathbf{Q}_t^{1/2} \boldsymbol{\eta}_t = \mathbf{L}_t \mathbf{G}_t^{1/2} \boldsymbol{\eta}_t \quad \Rightarrow \quad \mathbf{v}_t = \mathbf{G}_t^{1/2} \boldsymbol{\eta}_t.$$

Example : for $K+1=3$,

$$\mathbf{L}_t = \begin{pmatrix} 1 & 0 & 0 \\ \ell_{21,t} & 1 & 0 \\ \ell_{31,t} & \ell_{32,t} & 1 \end{pmatrix}, \quad \mathbf{B}_t = \begin{pmatrix} 1 & 0 & 0 \\ -\beta_{21,t} & 1 & 0 \\ -\beta_{31,t} & -\beta_{32,t} & 1 \end{pmatrix}$$

Orthogonal factors

The vector $\mathbf{v}_t$ can be interpreted as a vector of **orthogonal factors**, whose conditional covariance matrix

$$\mathbf{G}_t = V_{t-1}(\mathbf{v}_t)$$

is therefore **diagonal** (but not necessarily constant over time).

Thus

$$V_{t-1}(\mathbf{z}_t) := \mathbf{Q}_t = \mathbf{L}_t \mathbf{G}_t \mathbf{L}_t'$$

This is the Cholesky decomposition of the matrix $\mathbf{Q}_t$.

Parametrization of the dynamics through the β 's

Modelling a covariance matrix $\mathbf{Q}_t$ as a function of the past components of $\mathbf{z}_t$ is **difficult** because of

- (a) the possibly high dimensionality of the problem,
- (b) the constraint that $\mathbf{Q}_t$ must be **positive definite**.

In the decomposition

$$\mathbf{Q}_t = \mathbf{L}_t \mathbf{G}_t \mathbf{L}_t'$$

the positive-definiteness is **automatically guaranteed** if the (diagonal) elements of $\mathbf{G}_t$ are positive.

One has only to model

- the elements of the diagonal matrix $\mathbf{G}_t$ (i.e. the volatilities of the **orthogonal factors** v_{jt});
- the coefficients $\ell_{ij,t}$'s, or equivalently the $\beta_{ij,t}$'s (without positivity constraints).

Example : $\mathbf{Q} = \mathbf{LGL}' = \mathbf{DRD}$, $K = 1$ (1 factor)

$\mathbf{z} = \begin{pmatrix} f \\ y \end{pmatrix}$. In the second decomposition, $\mathbf{R}$ is the correlation matrix and $\mathbf{D}$ a diagonal matrix (with standard deviations on the diagonal). Letting

$$g_1 = \sigma_1^2 = \text{var}(f), \quad y = \beta_{21}f + v_2, \quad g_2 = \text{var}(v_2), \quad \sigma_2^2 = \text{var}(y)$$

and $\rho = \text{cor}(f, y)$, we have

$$\mathbf{L} = \begin{bmatrix} 1 & 0 \\ \beta_{21} & 1 \end{bmatrix}, \quad \mathbf{G} = \begin{bmatrix} g_1 & 0 \\ 0 & g_2 \end{bmatrix}, \quad \mathbf{D} = \begin{bmatrix} \sigma_1 & 0 \\ 0 & \sigma_2 \end{bmatrix}, \quad \mathbf{R} = \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix},$$

$$\mathbf{Q} = \begin{bmatrix} g_1 & \beta_{21}g_1 \\ \beta_{21}g_1 & \beta_{21}^2g_1 + g_2 \end{bmatrix} = \begin{bmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{bmatrix}.$$

Positivity constraints : $g_1 > 0$ and $g_2 > 0$ or $\sigma_1^2 > 0$, $\sigma_2^2 > 0$ and $\rho^2 < 1$.

Example : $\mathbf{Q} = \mathbf{LGL}'$, $\mathbf{Q}^{-1} = \mathbf{B}'\mathbf{G}^{-1}\mathbf{B}$, $K = 2$ (2 factors)

$$\mathbf{L} = \begin{bmatrix} 1 & 0 & 0 \\ \ell_{21} & 1 & 0 \\ \ell_{31} & \ell_{32} & 1 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 1 & 0 & 0 \\ -\beta_{21} & 1 & 0 \\ -\beta_{31} & -\beta_{32} & 1 \end{bmatrix}, \quad \mathbf{G} = \begin{bmatrix} g_1 & 0 & 0 \\ 0 & g_2 & 0 \\ 0 & 0 & g_3 \end{bmatrix},$$

$$\mathbf{Q} = \begin{bmatrix} g_1 & \ell_{21}g_1 & \ell_{31}g_1 \\ \ell_{21}g_1 & \ell_{21}^2g_1 + g_2 & \ell_{21}\ell_{31}g_1 + \ell_{32}g_2 \\ \ell_{31}g_1 & \ell_{21}\ell_{31}g_1 + \ell_{32}g_2 & \ell_{31}^2g_1 + \ell_{32}^2g_2 + g_3 \end{bmatrix}, \quad \text{with } g_1, g_2, g_3 > 0.$$

Remark : In $\mathbf{Q} = \mathbf{DRD}$ the constraints on the elements of $\mathbf{R}$ are

$$-1 \leq \rho_{ij} \leq 1, \quad \rho_{12}^2 + \rho_{13}^2 + \rho_{23}^2 - 2\rho_{12}\rho_{13}\rho_{23} \leq 1.$$

In $\mathbf{Q} = \mathbf{LGL}'$ there is **no constraint** on the ℓ_{ij} 's.

Parametrization of the conditional variances

Let $\mathbf{g}_t = (g_{1t}, \dots, g_{K+1,t})'$ the vector of diagonal elements of $\mathbf{G}_t$.

For instance, one can take a

- GARCH(1,1) model :

$$g_{it} = \omega_i + \alpha_i v_{i,t-1}^2 + \gamma_i g_{i,t-1}$$

- More complex volatility :
(spillovers between the orthogonal factors) :

$$g_{it} = \omega_i + \sum_{k=1}^i \alpha_{ik} v_{k,t-1}^2 + \gamma_i g_{i,t-1}$$

where all the coefficients are constrained to be positive.

Parametrization of the dynamics of the conditional β 's

These coefficients are not positively constrained.

For instance, one can take, for $i \geq j$

$$\beta_{ij,t} = \omega_{ij} + \alpha_{ij}v_{i,t-1} + \gamma_{ij}\beta_{ij,t-1}$$

or

$$\beta_{ij,t} = \omega_{ij} + \sum_{k=1}^i \alpha_{ijk}v_{k,t-1} + \gamma_{ij}\beta_{ij,t-1}$$

More generally

$$\boldsymbol{\beta}_t = \mathbf{c}(\mathbf{v}_{t-1}, \dots, \mathbf{v}_{t-p}, \mathbf{g}_{t-1}, \dots, \mathbf{g}_{t-q}) + \sum_{j=1}^r \mathbf{A}_r \boldsymbol{\beta}_{t-r}$$

where $\boldsymbol{\beta}$ is the vector of β_{ij} , $i = 1, \dots, K+1, j = 1, \dots, K$.

Stationarity conditions

The Cholesky decomposition of the matrix $\mathbf{Q}_t$ allows to derive **stationarity conditions** in which the elements of $\mathbf{G}_t$ and the betas can be handled separately.

For instance, if the g_{it} are modelled with GARCH, with

$$v_{it} = \sqrt{g_{it}}\eta_{it}, \quad g_{it} = \omega_i + \alpha_i v_{i,t-1}^2 + \gamma_i g_{i,t-1}$$

the usual stationarity conditions (for univariate GARCH) apply.

Then, if

$$\beta_{ij,t} = \omega_{ij} + \alpha_{ij} v_{i,t-1} + \gamma_{ij} \beta_{ij,t-1}$$

the constraint $|\gamma_{ij}| < 1$ ensures the stationarity of the $\beta_{ij,t}$.

Estimation

For a given parametric modeling, the model can be estimated by

① Full QML (Quasi-Maximum Likelihood)

based on $\mathbf{z}_t = \mathbf{Q}_t(\boldsymbol{\vartheta}_0)\boldsymbol{\eta}_t$ under the decomposition

$$\mathbf{Q}_t(\boldsymbol{\vartheta}_0) = \mathbf{L}_t(\boldsymbol{\vartheta}_0)\mathbf{G}_t(\boldsymbol{\vartheta}_0)\mathbf{L}_t'(\boldsymbol{\vartheta}_0)$$

② Multi-step QML

exploiting the hierarchical structure of the model.

One can estimate the triangular system equation-by-equation (EbE)

Full QML

A Gaussian QML estimator of $\boldsymbol{\vartheta}_0$ is obtained as

$$\hat{\boldsymbol{\vartheta}}_n = \arg \min_{\boldsymbol{\vartheta} \in \Theta} \sum_{t=1}^n q_t(\boldsymbol{\vartheta}),$$

where

$$\begin{aligned} q_t(\boldsymbol{\vartheta}) &= \mathbf{z}_t' \mathbf{Q}_t^{-1}(\boldsymbol{\vartheta}) \mathbf{z}_t + \log |\mathbf{Q}_t(\boldsymbol{\vartheta})| \\ &= \mathbf{z}_t' \mathbf{B}_t'(\boldsymbol{\vartheta}) \mathbf{G}_t^{-1}(\boldsymbol{\vartheta}) \mathbf{B}_t(\boldsymbol{\vartheta}) \mathbf{z}_t + \sum_{i=1}^{K+1} \log g_{it}(\boldsymbol{\vartheta}). \end{aligned}$$

Hence, it is not necessary to invert matrices ($\mathbf{G}_t(\boldsymbol{\vartheta})$ being diagonal) to compute the QML estimator. This is an attractive feature of this class of models.

Equation-by-Equation (EbE) estimator

Exploits the triangular form of the model.

For instance consider the system (with the true parameter) :
for $i = 2, \dots, K$,

$$f_{it} = \sum_{j=1}^{i-1} \beta_{ij,t} f_{jt} + v_{it}, \quad v_{it} = \sqrt{g_{it}} \eta_{it},$$

$$g_{it} = \omega_{0i} + \gamma_{0i} v_{i,t-1}^2 + b_{0i} g_{i,t-1}, \quad \beta_{ij,t} = \omega_{0,ij} + \tau_{0,ij} v_{i,t-1} + c_{0,ij} \beta_{ij,t-1}, \quad i > j.$$

Set

$$\boldsymbol{\vartheta}^{(i)} = (\omega_i, \gamma_i, b_i), \quad \boldsymbol{\varphi}^{(ij)} = (\omega_{ij}, \tau_{ij}, c_{ij})$$

We can compute recursively :

$$v_{1t} = f_{1t}, \\ g_{1t}(\boldsymbol{\vartheta}^{(1)}), \quad \beta_{21,t}(\boldsymbol{\varphi}^{(21)}, \boldsymbol{\vartheta}^{(2)}), \quad v_{2t}(\boldsymbol{\varphi}^{(21)}, \boldsymbol{\vartheta}^{(2)}), \quad g_{2t}(\boldsymbol{\vartheta}^{(2)}), \quad \dots$$

Equation-by-Equation (EbE) estimator, first step

Estimation of the volatility of the first factor

Consider the triangular model. In a **first step**, the true parameter $\boldsymbol{\vartheta}_0^{(1)} = (\omega_{01}, \gamma_{01}, b_{01})$ is estimated by

$$\hat{\boldsymbol{\vartheta}}_n^{(1)} = \arg \min_{\boldsymbol{\vartheta}^{(1)} \in \Theta^{(1)}} \sum_{t=1}^n q_{1t}(\boldsymbol{\vartheta}^{(1)}),$$

where

$$q_{1t}(\boldsymbol{\vartheta}^{(1)}) = \frac{f_{1t}^2}{g_{1t}(\boldsymbol{\vartheta}^{(1)})} + \log g_{1t}(\boldsymbol{\vartheta}^{(1)}),$$

and $g_{1t}(\boldsymbol{\vartheta}^{(1)}) = \omega_1 + \gamma_1 f_{1,t-1}^2 + b_1 g_{1,t-1}(\boldsymbol{\vartheta}^{(1)})$.

EbE second step

Volatility of the 2nd orthogonal factor and the dynamics of the 1st beta

Let $\boldsymbol{\vartheta}_0^{(2)} = (\boldsymbol{\varphi}_0^{(2)}, \boldsymbol{\theta}_0^{(2)})$, where $\beta_{21,t} = \beta_{21,t}(\boldsymbol{\varphi}_0^{(2)})$ and $g_{2t} = g_{2t}(\boldsymbol{\vartheta}_0^{(2)})$.

Independently or in parallel to $\boldsymbol{\vartheta}_0^{(1)}$, one can estimate $\boldsymbol{\vartheta}_0^{(2)}$ by

$$\hat{\boldsymbol{\vartheta}}_n^{(2)} = \arg \min_{\boldsymbol{\vartheta}^{(2)} \in \Theta^{(2)}} \sum_{t=1}^n q_{2t}(\boldsymbol{\vartheta}^{(2)}),$$

where, for $t = 1, \dots, n$,

$$q_{2t}(\boldsymbol{\vartheta}^{(2)}) = \frac{v_{2t}^2(\boldsymbol{\varphi}^{(2)})}{g_{2t}(\boldsymbol{\vartheta}^{(2)})} + \log g_{2t}(\boldsymbol{\vartheta}^{(2)}),$$

$$g_{2t}(\boldsymbol{\vartheta}^{(2)}) = \omega_2 + \alpha_2 v_{2,t-1}^2(\boldsymbol{\varphi}^{(2)}) + b_2 g_{2,t-1}(\boldsymbol{\vartheta}^{(2)}),$$

$$v_{2t}(\boldsymbol{\varphi}^{(2)}) = f_{2t} - \beta_{21,t}(\boldsymbol{\varphi}^{(2)}) f_{1t},$$

$$\beta_{21,t}(\boldsymbol{\varphi}^{(2)}) = \omega_{21,t-1} + \tau_{21} v_{2,t-1}(\boldsymbol{\varphi}^{(2)}) + c_{21} \beta_{21,t-1}(\boldsymbol{\varphi}^{(2)}).$$

The remaining steps can be treated similarly.

Full QML vs. EbE

- ① If $K = 1$, the one-step full QML and the two-step EbE estimators are exactly the same.
- ② For $K \geq 2$, the two estimators are generally different.
- ③ The EbE is simpler, and not always less efficient than the full QMLE.

Data Generating process

$$\mathbf{z}_t = \begin{pmatrix} \mathbf{f}_t \\ y_t \end{pmatrix} = \mathbf{Q}_t^{1/2}(\boldsymbol{\vartheta}_0)\boldsymbol{\eta}_t,$$

$$\mathbf{Q}_t = \mathbf{L}_t \mathbf{G}_t \mathbf{L}_t',$$

$$\mathbf{v}_t = \mathbf{G}_t^{1/2} \boldsymbol{\eta}_t, \quad (\boldsymbol{\eta}_t) \stackrel{iid}{\sim} (\mathbf{0}, \mathbf{I}_m),$$

where, for $i = 1, \dots, K+1$ and $j < i$,

$$g_{i,t} = 0.1 + 0.1\nu_{i,t-1}^2 + 0.8g_{i,t-1},$$

$$\beta_{ij,t} = 0.1 + 0.2\nu_{i,t-1} + 0.8\beta_{ij,t-1},$$

$$t = 1, \dots, n.$$

Results for $K = 4$, $\boldsymbol{\eta}_t \sim \mathcal{N}(\mathbf{0}, I_5)$, 1000 replications

	FULL QML		EbE	
	BIAS	RMSE	BIAS	RMSE
n=1000				
ω	0.0160	0.0614	0.0190	0.0672
α	0.0030	0.0295	0.0036	0.0322
β	-0.0196	0.0804	-0.0230	0.0876
φ	0.0006	0.0121	0.0007	0.0132
τ	0.0009	0.0149	0.0011	0.0165
c	-0.0013	0.0230	-0.0016	0.0250
ALL	0.0000	0.0301	0.0000	0.0329
n=2000				
ω	0.0068	0.0320	0.0087	0.0370
α	0.0013	0.0185	0.0017	0.0217
β	-0.0082	0.0439	-0.0104	0.0510
φ	0.0001	0.0067	0.0002	0.0079
τ	0.0002	0.0088	0.0004	0.0102
c	-0.0003	0.0126	-0.0004	0.0148
ALL	0.0000	0.0167	0.0000	0.0195

$$g_{i,t} = \omega + \alpha v_{i,t-1}^2 + \beta g_{i,t-1} \text{ and } \beta_{ij,t} = \varphi + \tau v_{i,t-1} + c \beta_{ij,t-1}$$

Results for $K = 4$, $\eta_{it} \sim t_7$, 1000 replications

	FULL QML		EbE	
	BIAS	RMSE	BIAS	RMSE
n=1000				
ω	0.0179	0.0689	0.0225	0.0771
α	0.0039	0.0355	0.0057	0.0414
β	-0.0233	0.0919	-0.0293	0.1031
φ	0.0008	0.0135	0.0010	0.0144
τ	0.0010	0.0155	0.0012	0.0178
c	-0.0016	0.0245	-0.0018	0.0271
ALL	-0.0001	0.0337	-0.0001	0.0378
n=2000				
ω	0.0064	0.0337	0.0093	0.0416
α	0.0009	0.0215	0.0020	0.0266
β	-0.0080	0.0478	-0.0120	0.0588
φ	0.0002	0.0065	0.0003	0.0080
τ	0.0003	0.0083	0.0005	0.0101
c	-0.0005	0.0124	-0.0007	0.0150
ALL	-0.0001	0.0175	0.0000	0.0215

$$g_{i,t} = \omega + \alpha v_{i,t-1}^2 + \beta g_{i,t-1} \text{ and } \beta_{ij,t} = \varphi + \tau v_{i,t-1} + c \beta_{ij,t-1}$$

Asset Pricing for Industry Portfolios

- We consider the 12 industry portfolios used by Engle (2016), examined in the context of the Fama French 3 factor model.
- The three factors are : *MKT* (Market factor = excess log-returns of the SP500), *SMB* (small minus big size factor) and *HML* (high minus low value factor)
- Data are from Ken French's web site and cover the period 1994-2016.

Competing models

Let $\mathbf{z}_t = (\mathbf{f}_t', y_t)'$ with $\mathbf{f}_t = (MKT_t, SMB_t, HML_t)'$ and $y_t = r_{kt}$.

Hedging strategy :

$$E_{t-1}(r_{kt} | \mathbf{x}_t) = \beta_{k,MKT,t} MKT_t + \beta_{k,SMB,t} SMB_t + \beta_{k,HML,t} HML_t.$$

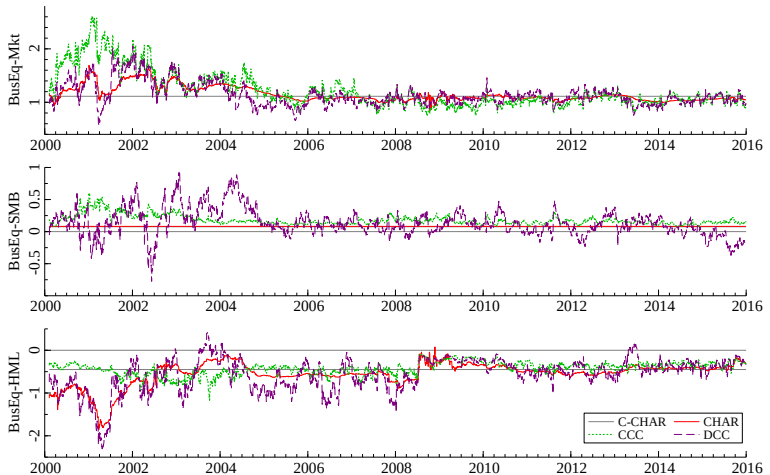
Competing models :

- ① CCC-GARCH(1,1)
- ② DCC-GARCH(1,1)
- ③ CHAR with constant betas
- ④ CHAR with time varying betas $\beta_{ij,t} = \omega_{ij} + \tau_{ij} v_{i,t-1} v_{j,t-1} + c_{ij} \beta_{ij,t-1}$

CHAR model on $(MKT_t, SMB_t, HML_t, r_{kt})'$ or $(MKT_t, HML_t, SMB_t, r_{kt})'$ by minimizing AIC (or equivalently BIC).

Conditional betas for Business Equipment

Computers, softwares, electronic equipment



Comparison of models' predictions

The usual practice in the comparison of models for prediction is to

- Estimate the models on a training sample (n_0 observations)
- Compare the predictions "out of sample" (n_1 observations)

Let $\hat{Y}_{i,t}$ the forecast of Y_t by a given model i .

The quadratic loss is : $L_{it} = (\hat{Y}_{i,t} - Y_t)^2$

The root-MSE of prediction of model i is :

$$\text{RMSE}_i = \sqrt{\frac{1}{n_1} \sum_{t=n_0+1}^{n_0+n_1} L_{it}^2}.$$

Model i is better than model j if $\text{RMSE}_i < \text{RMSE}_j$ but how **significantly** better ?

Diebold-Mariano test

We would like to test the null hypothesis

$$\mathbf{H}_0: EL_{1t} = EL_{2t} \quad \Leftrightarrow \quad \mathbf{H}_0: Ed_t = 0$$

where $d_t = L_{1t} - L_{2t}$, versus the alternative that the two forecasts have two different level of accuracy

$$\mathbf{H}_1: Ed_t \neq 0$$

Consider a sequence of loss differences, $d_1, \dots, d_T$ and their sample mean $\bar{d}_T = \frac{1}{T} \sum_{t=1}^T d_t$.

Diebold-Mariano test

It can be shown that under some **regularity conditions** (2nd-order stationarity+ short memory), we have under $\mathbf{H}_0$,

$$\sqrt{T}\bar{d}_T \overset{d}{\rightsquigarrow} \mathcal{N}\left(0, \omega^2 := \sum_{k=-\infty}^{\infty} \gamma(k)\right)$$

where $(\gamma(k))$ is the sequence of autocovariances of (d_t) .

Thus under $\mathbf{H}_0$,

$$\sqrt{T} \frac{\bar{d}_T}{\hat{\omega}} \overset{d}{\rightsquigarrow} \mathcal{N}(0, 1)$$

where $\hat{\omega}$ is any consistent estimator of ω .

A test of $\mathbf{H}_0$, at the asymptotic level $\alpha \in (0, 1)$, is thus

$$C = \left\{ T \frac{\bar{d}_T^2}{\hat{\omega}^2} > \chi_1^2(1 - \alpha) \right\}.$$

Diebold-Mariano test

Remark : A consistent estimator of the "long run" variance ω^2 is obtained by **weighting the empirical autocovariances** $\hat{\gamma}(k)$ of the d_t (using a weight close to 1 when k is small, and close to 0 when k is large) :

$$\hat{\omega}^2 = \sum_{k=-\infty}^{\infty} K(kh_T) \hat{\gamma}(k),$$

where K is a kernel (which means that $K(x) \geq 0$ and $\int K(x)dx = 1$) and h_T is a bandwidth parameter ($h_T \rightarrow 0$ when $T \rightarrow \infty$).

Standard kernels : $K(x) = \mathbf{1}_{[-0.5, 0.5]}(x)$; $K(x) = (1 - |x|) \mathbf{1}_{[-1, 1]}(x)$ (Bartlett kernel), Parzen, Tukey-Hanning, etc *

The naive estimator $\sum_{k=-\infty}^{\infty} \hat{\gamma}(k)$ is **inconsistent**.

*. See M.B. Priestley, Spectral Analysis and Time Series, Vols. 1 and 2, Academic Press, New York, 1981.

For a set of competitive models : Models Confidence Sets (MCS)

Hansen, Lunde and Nason, 2011

Idea : determine a set of models

$$\mathcal{M}^* = \{\text{superior models belonging to } \mathcal{M}\}.$$

The MCS procedure produces a set $\mathcal{M}_{1-\alpha}^*$: a collection of models built to contain the best ones with a confidence level $(1 - \alpha)\%$

The relative performance of model j w.r.t. model i is $d_{ij,t} = L_{i,t} - L_{j,t}$.

Assuming $\mu_{ij} = E(d_{ij,t})$ exists

$$\mathcal{M}^* = \{i \in \mathcal{M}; \quad \mu_{ij} \leq 0 \quad \forall j \in \mathcal{M}\}$$

Models Confidence Sets (MCS)

Hansen, Lunde and Nason, 2011

MCS is based on a sequential procedure for determining $\mathcal{M}_{1-\alpha}^*$:

- Start from $\mathcal{M}$: test at level α the null $\mathbf{H}_{0,\mathcal{M}} : \mu_{ij} = 0, \forall i, j$.
- If the null is rejected, the procedure eliminates a (bad) model from $\mathcal{M}$ and repeats **until the null of no difference** cannot be rejected.

The set $\mathcal{M}_{1-\alpha}^*$ is the set of remaining models.

Models Confidence Sets (MCS)

Hansen, Lunde and Nason, 2011

Tests are based on the statistics

$$t_{ij} = \frac{\bar{d}_{ij}}{\sqrt{\hat{V}(d_{ij})}}, \quad \text{where} \quad \bar{d}_{ij} = \frac{1}{T} \sum_{t=1}^T d_{ij,t}$$

Test statistics of $\mathbf{H}_{0, \mathcal{M}_1} : \mu_{ij} = 0, \forall i, j$, for a set of models $\mathcal{M}_1$:

$$S_{\mathcal{M}_1} = \max_{i,j \in \mathcal{M}_1} |t_{ij}|$$

- This statistics only depends on the model that has the greatest relative loss.
- The asymptotic distribution of $S_{\mathcal{M}_1}$ is unknown (because the t_{ij} are dependent).
- The MCS procedure uses the bootstrap to evaluate the distribution of $S_{\mathcal{M}_1}$.

Model Confidence Set test

Hansen, Lunde and Nason, 2011

	C-CHAR	CHAR	CCC	DCC
BusEq		✓		
Chems		✓		
Durbl		✓		
Enrgy		✓		✓
Hlth		✓		✓
Manuf		✓		
Money		✓		✓
NoDur		✓		
Other		✓		
Shops		✓		
Telcm		✓		
Utils		✓		

Models highlighted with the symbol ✓ are contained in the model confidence set using a MSE loss function. The significance level for the MCS is set to 20%, and 10,000 bootstrap samples (with a block length of 5 observations).

End of the Course

Conditional distributions of the components of a Gaussian vector

Let

$$\begin{pmatrix} \mathbf{X} \\ \mathbf{Y} \end{pmatrix} \sim \mathcal{N} \left(\begin{pmatrix} \boldsymbol{\mu}_X \\ \boldsymbol{\mu}_Y \end{pmatrix}, \begin{pmatrix} \boldsymbol{\Sigma}_{XX} & \boldsymbol{\Sigma}_{XY} \\ \boldsymbol{\Sigma}_{YX} & \boldsymbol{\Sigma}_{YY} \end{pmatrix} \right).$$

Then the distribution of $\mathbf{X}$ conditional on $\mathbf{Y} = \mathbf{y}$ is

$$\mathcal{N} \left(\boldsymbol{\mu}_X + \boldsymbol{\Sigma}_{XY} \boldsymbol{\Sigma}_{YY}^{-1} (\mathbf{y} - \boldsymbol{\mu}_Y), \quad \boldsymbol{\Sigma}_{XX} - \boldsymbol{\Sigma}_{XY} \boldsymbol{\Sigma}_{YY}^{-1} \boldsymbol{\Sigma}_{YX} \right).$$

◀ Return

Vector GARCH model

$\mathbf{H}_t$: conditional variance of a process $(\boldsymbol{\epsilon}_t)$.

vech (vector half) operator : takes a symmetric $d \times d$ matrix and stacks the lower triangular half into a vector of length $d(d+1)/2$.

$$\boldsymbol{\epsilon}_t = \mathbf{H}_t^{1/2} \boldsymbol{\eta}_t, \quad \mathbf{H}_t \text{ positive definite, } (\boldsymbol{\eta}_t) \text{ iid } (\mathbf{0}, \mathbf{I})$$

$$\text{vech}(\mathbf{H}_t) = \boldsymbol{\omega} + \sum_{i=1}^q \mathbf{A}^{(i)} \text{vech}(\boldsymbol{\epsilon}_{t-i} \boldsymbol{\epsilon}_{t-i}') + \sum_{j=1}^p \mathbf{B}^{(j)} \text{vech}(\mathbf{H}_{t-j})$$

- The most direct generalization of univariate GARCH
- Positivity conditions are difficult to obtain
- No explicit stationarity conditions

BEKK-GARCH model

Engle and Kroner (1995), Comte and Lieberman (2003)

$$\left\{ \begin{array}{l} \boldsymbol{\epsilon}_t = \mathbf{H}_t^{1/2} \boldsymbol{\eta}_t, \quad (\boldsymbol{\eta}_t) \text{ iid } (\mathbf{0}, \mathbf{I}) \\ \mathbf{H}_t = \boldsymbol{\Omega} + \sum_{i=1}^q \sum_{k=1}^K A_{ik} \boldsymbol{\epsilon}_{t-i} \boldsymbol{\epsilon}_{t-i}' A_{ik}' + \sum_{j=1}^p \sum_{k=1}^K B_{jk} \mathbf{H}_{t-j} B_{jk}' \end{array} \right.$$

- Coefficients of a BEKK representation are difficult to interpret
- Positivity conditions are simple. Identifiability of a BEKK representation requires additional constraints.
- Stationarity conditions exist (Boussama, Fuchs, Stelzer, 2011) but no explicit solution can be exhibited

Constant Conditional Correlation (CCC) model

Bollerslev (1990); Extended CCC by Jeantheau (1998)

$$\underline{h}_t = \begin{pmatrix} h_{11,t} \\ \vdots \\ h_{mm,t} \end{pmatrix}, \quad D_t = \text{diag}(h_{11,t}^{1/2}, \dots, h_{mm,t}^{1/2}), \quad \underline{\epsilon}_t = \begin{pmatrix} \epsilon_{1t}^2 \\ \vdots \\ \epsilon_{mt}^2 \end{pmatrix}.$$

$$\begin{cases} \underline{\epsilon}_t = H_t^{1/2} \eta_t, & H_t = D_t \mathbf{R} D_t, \quad \mathbf{R} : \text{correlation matrix} \\ \underline{h}_t = \omega + \sum_{i=1}^q \mathbf{A}_i \underline{\epsilon}_{t-i} + \sum_{j=1}^p \mathbf{B}_j \underline{h}_{t-j} \end{cases}$$

- Simple conditions ensuring the positive definiteness of H_t .
- Explicit stationarity condition (of the form $\gamma < 0 \dots$)
- The assumption of CCC can be too restrictive

Dynamic Conditional Correlation (DCC) model

Engle (2002)

$$\mathbf{H}_t = \mathbf{D}_t \mathbf{R}_t \mathbf{D}_t, \quad \mathbf{R}_t = (\text{diag } \mathbf{Q}_t)^{-1/2} \mathbf{Q}_t (\text{diag } \mathbf{Q}_t)^{-1/2},$$

where $\boldsymbol{\eta}_t^* = \mathbf{D}_t^{-1} \boldsymbol{\epsilon}_t$ and

$$\mathbf{Q}_t = (1 - \alpha - \beta) \mathbf{S} + \alpha \boldsymbol{\eta}_{t-1}^* \boldsymbol{\eta}_{t-1}^{*'} + \beta \mathbf{Q}_{t-1},$$

where $\alpha, \beta \geq 0, \alpha + \beta < 1$, $\mathbf{S}$ is a correlation matrix

- The existence of strictly stationary solution is a complex issue
- No asymptotic theory of estimation exists
- Incorrect interpretation of $\mathbf{S}$ as $\text{Var}(\boldsymbol{\eta}_t^*)$ and $\mathbf{Q}_t$ as $\text{Var}_{t-1}(\boldsymbol{\eta}_t^*)$.

Dynamic Conditional Correlation (DCC) model

Corrected DCC (Aielli (2013))

$$\mathbf{H}_t = \mathbf{D}_t \mathbf{R}_t \mathbf{D}_t$$

$$\mathbf{R}_t = (\text{diag } \mathbf{Q}_t)^{-1/2} \mathbf{Q}_t (\text{diag } \mathbf{Q}_t)^{-1/2},$$

where

$$\mathbf{Q}_t = (1 - \alpha - \beta) \mathbf{S} + \alpha \mathbf{Q}_{t-1}^{*1/2} \boldsymbol{\eta}_{t-1}^* \boldsymbol{\eta}_{t-1}^{*'} \mathbf{Q}_{t-1}^{*1/2} + \beta \mathbf{Q}_{t-1},$$

where $\mathbf{Q}_t^* = \text{diag}(\mathbf{Q}_t)$.

- Identifiability constraint : $\text{diag}(\mathbf{S}) = \mathbf{I}_m$.
- Parsimony but the $m(m-1)/2$ conditional correlations have the same dynamic structure.